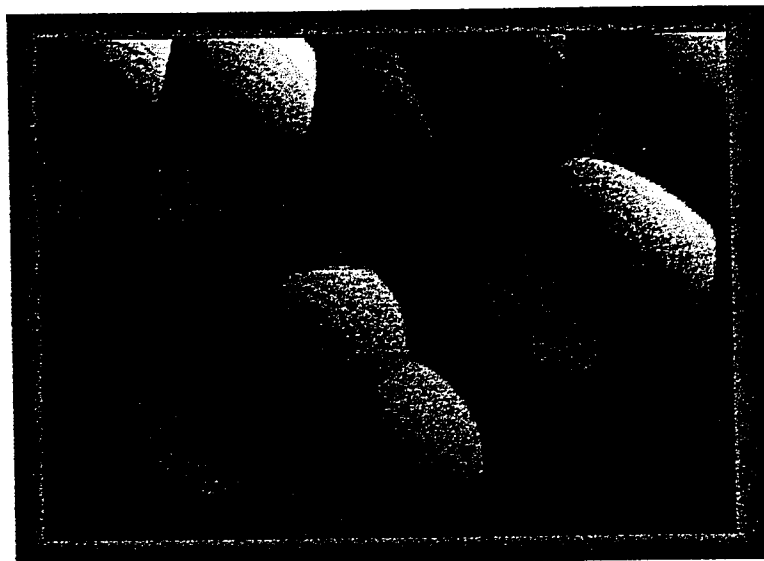


BioSize™

Molecular Weight Determination Software with SimGel™ Viewing Utility



Instruction Manual

For Technical Service
Call your Local Bio-Rad Office or
in the U.S. Call **1-800-4BIORAD**
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BioSize™ and SimGel™ require Windows 3.1 or Windows 95 and BioFocus Integrator version 5.0 to operate.

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The Basics

Introduction

BioSize version 5.0 is a software program that estimates the molecular weights of sample proteins based on the migration times of a series of standard proteins. The information used by this software is extracted from files analyzed and stored in the BioFocus Integration database. This software also incorporates Bio-Rad's SimGel utility for converting and viewing electropherograms as simulated gel patterns.

Using this software is a two step process:

1. A standard curve is defined by entering migration times and molecular weights from the integration report of a capillary electrophoresis run of standard proteins.
2. After the standard curve is defined, a sample protein is analyzed by selecting its integration report file and running the BioSize analysis feature.

The software allows easy batch processing with the ability to use multiple standard runs to correct for run-to-run variations in migration time. In addition, if an internal reference was included in the standard and sample solutions, its migration time in the integration reports can be used to compensate for any day-to-day and capillary-to-capillary variations.

The Main Screen

NOTE: *BioFocus Integration software version 5.01 (or higher) must be installed for BioSize to operate correctly.*

The **Main** screen controls how data is entered and analyzed. The file selection controls are on the left side of the screen. Each file in the BioFocus Integration database is listed. Files in the integration database that have not been integrated are grayed out. You may view these files in SimGel mode, but you must integrate them before you can estimate the molecular weight.

Bio-Rad Bio Size

File Edit View Data Help Window

Select/Define a Standard

File Selection

Date	Time	Run Name
08-14-95	1:39 PM	5klad001:500-4000 ladder:00
08-14-95	1:33 PM	acids001:3 acids:0001
01-15-95	6:46 PM	days 006:standards 10 ppa
12-01-92	11:33 AM	exstd001:Level 1:0003
12-01-92	11:23 AM	exstd002:Level 2:0004
12-01-92	11:12 AM	exstd003:Level 3:0005
10-28-93	5:43 PM	sdsma010:Protein MW stds,
10-28-93	7:21 PM	sdsma015:IgG1, reduced:000
08-15-95	10:23 AM	dnalr001:DNA Ladder-Low ra
12-01-92	11:33 AM	exstd001:Level 1:0009
12-01-92	11:23 AM	exstd002:Level 2:0010
12-01-92	11:12 AM	exstd003:Level 3:0011
12-01-92	11:43 AM	exstd004:HIGH:0012
08-15-95	10:23 AM	dnalr001:DNA Ladder-Low 12

Files to Analyze

File Name...	Peaks	Status

Operations

AutoPrint Bracket Analyze

Figure 1. The Main Screen

Parts of the Main Screen

On the left side of the main screen is the **File Selection** list box. It lists those files that are in the BioFocus Integration database and are available for analysis using BioSize or SimGel.

The **Files to Analyze** list box is located on the right side of the screen. After a standard has been defined, double-clicking on a file from the **File Selection** list box, will cause it to be added to the **Files to Analyze** list box.

Across the top of the screen is a toolbar with eight buttons. The buttons on the left side are used to control software functions, and the ones on the right are used for entering or erasing files from the Files to Analyze list box. In order from left to right, the buttons are **Standard**, **SimGel**, **Mode**, **Help**, **Erase Row**, **Insert Row**, **Erase All**, and **Exit**. The function of each button is explained in the appropriate sections of this manual.

Select the **Standard** button to activate the **Standard Definition** screen. If no standard has been defined, double-clicking on a file in the **File Selection** list box will also open the **Standard Definition** screen.

The **Mode** button is used to change from BioSize mode to SimGel mode. BioSize mode is used to estimate molecular weight, and SimGel is used to view electropherograms and simulated gel patterns.

Defining the Standard

Standard Definition

Prior to a BioSize analysis, a standard must be defined. The standard definition is used to relate molecular weight to migration time. After a protein standard is analyzed by capillary electrophoresis and the result integrated, the integration report can be used to define the standard for BioSize. The molecular weight range of the standard proteins should include the molecular weight range of the sample proteins. For example, the CE-SDS Protein Size Standard from Bio-Rad has proteins that range from 14 kD to 200 kD in molecular weight; therefore, if you use the Bio-Rad protein standard, your sample proteins should be within this range.

There are two ways to define a standard. The first method involves entering a user-defined standard where each entry is entered manually. Click on the **Standard** button. The **Standard Definition** screen will come up with some entries already in place; these can be typed over. Proceed to type in the times and molecular weights desired.

The second method involves running a protein size standard and using the data from that run to automatically fill in the standard definition. To define a standard based on an electrophoresis run, select that run from the **File Selection** list box on the main screen by clicking once and then clicking the **Standard** button. The **Standard Definition** screen will appear with the times and weights already entered. By default, the screen uses the molecular weights from the Bio-Rad CE-SDS Protein Size Standard. If you have used a different standard, the correct molecular weights must be typed over the default values. See “Editing the Standard Definition Screen” on page 9 for more information on this procedure.

Defining a Standard from a Report

Opening the Report File

Once an electropherogram has been analyzed with the BioFocus Integrator, the migration times can be used to define the Standard. Report files are stored in the

integrator database. To use a file as a standard, select the file by double-clicking on it.

The Standard Definition screen will appear with the retention time and molecular weights filled in. When you select a report file before clicking on the standard button, the migration times are automatically entered.

Time	Mol.Weight
1.58	14400
5.97	21500
6.48	31000
7.06	45000
8.05	66200
8.67	97400
9.22	116250
9.56	200000
10.86	

Figure 3. Standard From a Report

Figure 3 is an example of the **Standard Definition** screen that was filled using migration times from a report file and molecular weights based on the Bio-Rad CE-SDS Protein Size Standard (Catalog number 148-2015). When you select a report file and then click on the **Standard** button, the table is automatically filled in with migration times from the selected report file and molecular weight values associated with the Bio-Rad protein standard.

The reason Figure 3 contains an empty molecular weight cell for the migration time at 10.86 min. is that the report file selected contains an internal standard at migration time 1.58 minutes. This produces nine peaks in the file, while the protein standard contains only eight proteins. The next section explains editing the standard definition screen and working with internal references.

Editing the Standard Definition Screen

Basic Editing

***Hint:** The molecular weights from Bio-Rad's protein standard can automatically be entered by selecting the Bio-Rad Standard option on the Edit pull-down menu.*

Entering Values

Type in the migration times and corresponding molecular weights in the **Standard Definition** screen.

Enter "REF" for the molecular weight of the internal reference peak (See "Internal Reference" on page 10 for more information). If there are peaks in your standard that are impurities, or if you do not want them included in the molecular weight determination, leave the molecular weight entries blank for those peaks, or delete the migration time for that peak from the **Standard Definition** screen. Any peak without a corresponding molecular weight entry will be ignored when calculations are performed.

However, in the case of batch processing, special precautions must be taken in the definition of a standard. See "Standard for Batch Processing" on page 10 for more information.

Inserting and Deleting Entries

To insert a row in either the time or weight columns, click the mouse cursor in the cell you want blank, then choose **Insert Row** from the **Edit** pull-down menu. The selected entry and every entry below that point will be shifted down leaving room to insert a new entry. To delete an entry, select the entry with the mouse and choose **Delete Row** from the **Edit** pull-down menu.

To clear all entries in the define standard screen, choose **Delete All** from the **Edit** pull-down menu.

Internal Reference

Example: If a reference peak is designated at 2.5 minutes, all peak times will be divided by 2.5 minutes. A migration time of 5 minutes will have a relative migration time of 2.

Any peak can be designated as the internal reference by entering "REF" in the **Molecular Weight** column. Only one reference peak is allowed. Once a reference peak is designated, all calculations will be based on relative migration times.

The **Reference Window** determines the time range the software will search for a reference peak when it analyzes data. If a reference peak is not found, calculations will be performed without an internal reference, and the migration times will be absolute rather than relative. Unless you are using batch-mode processing, you will get an error indicating that an internal reference was not found.

If an internal reference was not designated, the value for **Reference Window** is ignored. If more than one peak falls within the **Reference Window**, the peak closest in migration time to the internal reference defined in the **Standard Definition** screen will automatically be used as the reference peak.

Standard for Batch Processing

Standard Without Internal Reference

Batch processing allows you to process multiple sample files with unattended operation. In this mode, a new standard definition can be periodically substituted for the one defined in the **Standard Definition** screen. To update this standard, peak order is used to correctly match the migration time with the molecular weight. This requires the number of migration times in the **Standard Definition** screen to be exactly the same as the number of peaks in the report file. See "Using Standard Runs in Analysis." on page 17 for more information.

The standard mixture may have peaks that are unknown or that should not be included in the molecular weight calculations. To reject these unwanted peaks, enter a time for the peak, but leave the molecular weight entry blank. In Figure 4, the standard mixture has 10 peaks, but the peaks at 1.55 and 9.37 are contaminants. By leaving those molecular weight entries blank, the peaks at 1.55 and 9.37 will not be used for any calculations.

Time	Mol. Weight
1.55	
5.85	14400
6.35	21500
6.92	31000
7.88	45000
8.5	66200
9.03	97400
9.37	
9.96	116250
10.64	200000

Figure 4. Defining the Standard

*In figure 5, the **Reference Window** is set to ± 0.5 minutes. If a new standard run is selected during batch processing, the reference peak must be within 0.5 minutes of the designated reference time or an error will result*

Standard With Internal Reference

If there is an internal reference in your standard mixture, enter "REF" in the column for its molecular weight. Only one internal reference is allowed.

For example, in Figure 5, the standard mixture has a contaminate at 1.55 minutes and an internal reference at 5.85 minutes. For this example, the molecular weights will be calculated using eight peaks. Since the peak at 5.85 minutes is designated as a reference peak, all calculations will be based on relative migration time.

The **Reference Window** designates the time frame for determining which peak is a reference during batch processing.

Time	Mol.Weight
1.55	
5.85	REF
6.35	14400
6.92	21500
7.88	31000
8.5	45000
9.03	66200
9.37	97400
9.96	116250
10.64	200000

Ref. Window: + / - 0.5 Min.

Buttons: Save, Open, View Standard, OK

Figure 5. Defining a Reference Peak

Managing Your Standard Definition

Saving a Standard Definition

Once your standard is defined, you can save it to the hard disk by selecting the **Save** button. BioSize Standard Definition files saved using this button are given the file extension STD by default; however, any legitimate DOS name will work.

BioSize Standard Definition files are ASCII text files that can be edited with any word processor. They must have the heading "Standard Definition File", and be followed by the list of migration times and molecular weights. The times must be separated from the weights by a single comma, and each entry must be on a separate line.

Opening a Saved Standard Definition

Select the **Open** button to use a previously defined standard. A list of BioSize Standard Definition files will be present. Select the file you want and choose OK. The **Standard Definition** screen will be filled in with the values from the file.

Viewing a Standard Calibration

To see what the standard curve looks like, choose the **View Standard** button on the **Standard Definition** screen. A **View Standard** window will open with a plot of log MW versus migration time or relative migration time.

On the right side of the screen is a scroll bar. Scrolling down a page will display the numeric results of the molecular weight calibration.

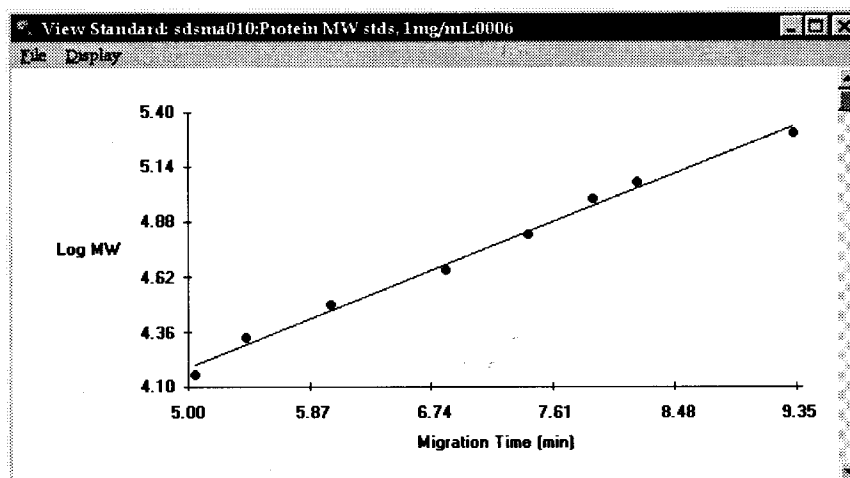


Figure 6A. View Standard Screen

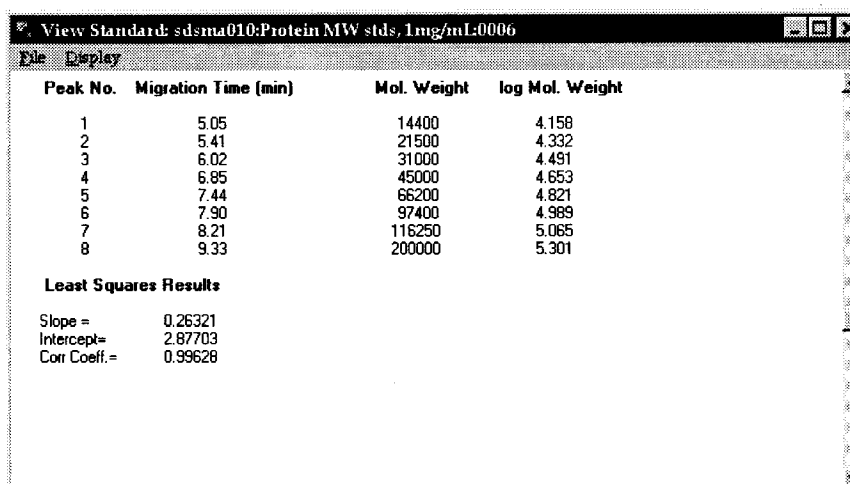


Figure 6B. View Standard Screen

Analyzing an Electropherogram

The Analyze Button

Once a standard is defined, the **Analyze** button can be used to analyze files. After defining a standard, the status display on the top of the screen will read "Select files to analyze".

Selecting the File

Select the file to analyze by double-clicking on the file in the File Selection box. The file will appear in the **Files to Analyze** box. You can add as many files to this list as you like.

Viewing the Report

Once a file is selected, click on the Analyze button. The **Report** screen will open and display the plot of the standard and unknown data (see Figure 7). If the default colors are being used, the standards are plotted in light blue and the unknowns are plotted in gray. A best-fit line is drawn and the samples are plotted against that line.

On the right side of the screen is a scroll bar. Scrolling down will display the numeric results of the molecular weight determination.

Exiting Report

To exit the report and return to the **Main** menu, select **Exit** from the **File** pull-down menu on the **Report** screen.

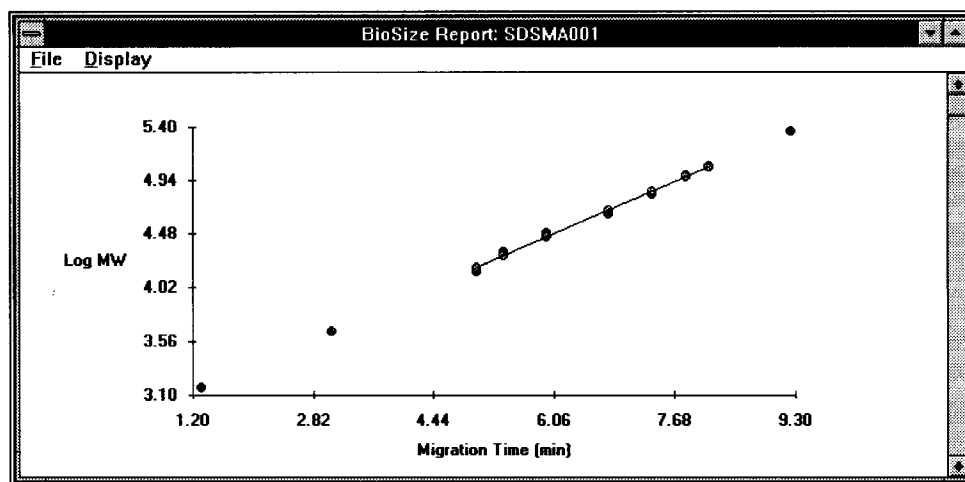


Figure 7. The Report Screen

Printing the Report

To print the report, select **Print** from the **File** pull-down menu on the **Report** screen. A Windows print dialog box will appear. Select the desired options and choose **OK**.

Report Format

The **Report Format** command is located on the **Display** pull-down menu. This command lets you select how the report is displayed. By selecting the appropriate option box, you can choose to print only the standards or sample points. Axis ranges, tick marks, and grid markings can also be selected. Once customized, you can save these settings as the default settings by clicking the "Save as Default" button.

Report Format

☒ Plot Standard Points ☐ y Grid

☒ Plot Sample Points ☐ x Grid

☐ Show MW Labels

☒ Autoscale Plot

Scaling Control

x max: 15 y max: 7

x min: 4.5 y min: 4

x tick every: 2.09 y tick every: .6

Save as Default **Done**

Figure 8. The Report Format Screen

Report Colors

The **Report Colors** command is located on the **Display** pull-down menu. This option allows you to select the colors used on the report. Each item on the graph can have a different color. The default colors highlight Out-of-Range points in light red, Standards in light blue, and Samples in gray. Once customized, you can save these settings as the default settings by clicking the “Save as Default” button.

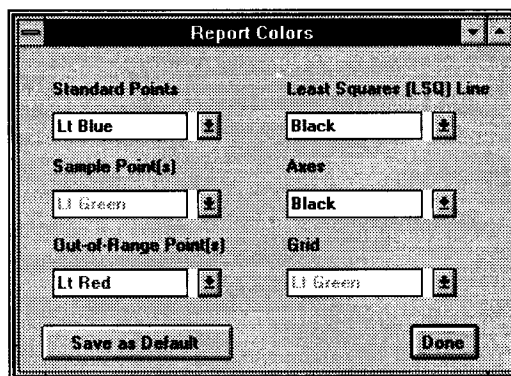


Figure 9. The Report Colors Screen

Out-of-Range Points are those that fall outside the molecular weight range defined by your standard. For example, if the smallest peak in your standard was 14 kD, than any peak calculated to be less than 14 kD will be out of range.

Least-Squares (LSQ) Line color refers to the color of the least-squares-fit line that is calculated from the defined standard.

The Files to Analyze Section

Using Standard Runs in Analysis.

Use standard runs in the Files to Analyze list box to correct for migration time variability.

It is common to re-run a standard after a number of sample runs to correct for variations in migration time over the total run time. To designate a run as a standard run, click the mouse on the table in the column marked **Status**. The text in the column will change to "STD".

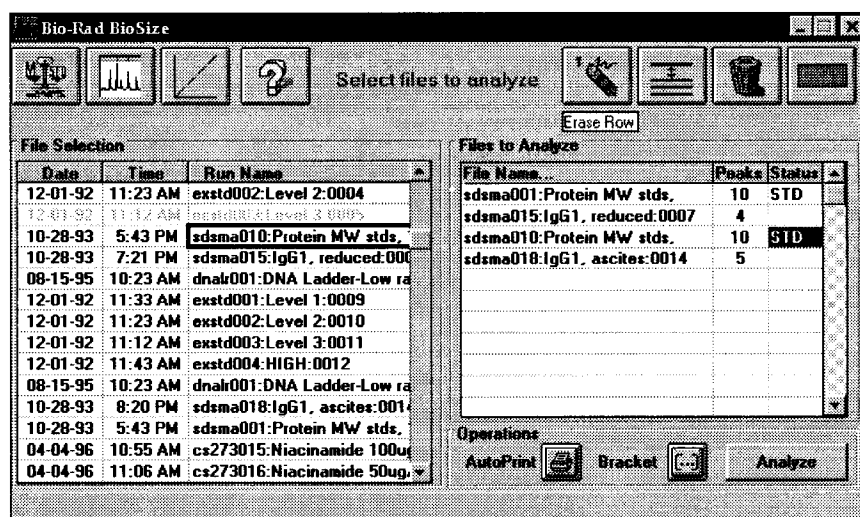


Figure 12. Using Standards in Batch Processing

In Figure 12, runs SDSMA001 and SDSMA010 are designated as standards. The run named SDSMA015 will be analyzed using the standard SDSMA001. This occurs because designating SDSMA001 as a standard in the **Files to Analyze** table causes BioSize to replace the migration times listed in the **Standard Definition** screen with those from run SDSMA001. Run SDSMA018 is analyzed using the standard run SDSMA010 because designating SDSMA010 as a standard in the **Files to Analyze** table causes the standard to be replaced with values from SDSMA010.

Each standard run you designate will replace the one you defined previously. All subsequent runs will be analyzed using this new standard definition.

An error will occur if you attempt to replace the defined standard with a new standard that has a different number of peaks.

To change the status of an entry in the **File to Analyze** table from standard to sample, click the mouse in that cell again and a blank cell will appear.

Each time the program encounters a run designated as a standard, the migration times for the defined standard will be replaced by the times from the designated run. Migration times are replaced exactly. The first peak in the designated standard run will replace the first peak in the defined standard. This means the designated standard run in the **Files to Analyze** table must have the same number of peaks as the standard you first defined. In other words, you cannot change from using one type of protein standard to a second type in the same analysis.

Once a batch analysis is complete, the standard definition reverts to that originally defined in the **Standard Definition** screen.

Averaging Standard Runs

***Bracket** buttons allow you to average standard migration times.*



Bracket button



Bracket - Enabled

Standard runs can be averaged by clicking on the **Bracket** button with the mouse.

To correctly bracket runs, you must designate at least two runs as standards in the **Files to Analyze** table. When the **Bracket** button is enabled, the migration times from the two standards will be averaged together to create a new standard. The sample runs between the two standard runs in the table will be analyzed based on the new averaged standard.

Example: In Figure 12, the run SDSMA015 would normally be analyzed using standard SDSMA001. If the **Bracket** button is enabled, it will be analyzed with the standard created from the average of SDSMA001 and SDSMA010. The run that follows the second standard in the average definition will be analyzed using only the data from the second standard since it is not part of a bracket analysis.

Printing Analysis

*When the **Print** button is selected, the entire table will be analyzed and printed.*



*The **Print** button*

When performing batch analysis, it is possible to automatically print the results. To do this, depress the **Print** button prior to pushing the **Analyze** button. Then when the **Analyze** button is pushed, the sample files in the **Files to Analyze** table will be processed and the results automatically printed.

Release the **Print** button to disable the automatic printing feature.

SimGel Button

Introduction to SimGel

SimGel is a viewing utility that simulates slab-gel electrophoresis patterns from capillary electropherograms. Its goal is to provide users, whose backgrounds are in slab-gel electrophoresis, with an alternate but familiar view of their capillary electrophoresis patterns.

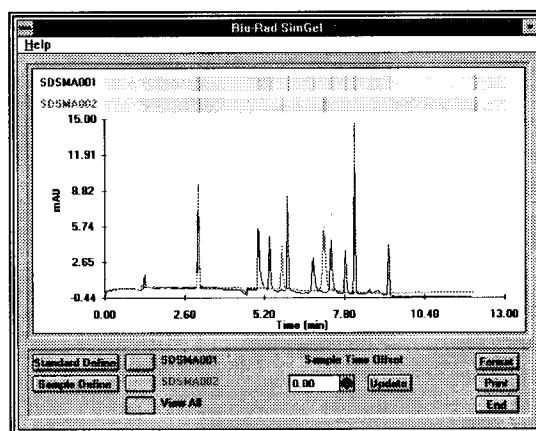
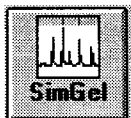


Figure 13. SimGel Display

Accessing SimGel



SimGel Button

The **SimGel** button launches the SimGel viewing utility and will display electropherograms and simulated slab-gel electrophoresis bands for the standard and the last sample file analyzed.

Alternatively, the software can be placed in SimGel mode by depressing the **Mode** button. Once in SimGel mode, the “Files to Analyze” list box is used to designate which files to view using SimGel.

Viewing a File

Selecting the File

The first step to plotting a SimGel is to select a file. Files are selected by clicking on the file in the **File Selection** box. Once a file is selected, click on the SimGel button.

Alternatively selecting a file by clicking on it with the right mouse button will start SimGel.

To view two files, depress the **Mode** button to place the software in SimGel mode. Double-click on the first file and it will appear in the "Files to Analyze" list box. Double-click on another file and it will appear in the "Files to Analyze" list box. Click on the **Analyze** button to see the simulated gel.

Zooming In on the Display

To zoom in on a display, point to the upper left corner of the region you would like to zoom into, and push the left mouse button down. Keeping the mouse button down, move the mouse and create a box around the area of interest. Release the mouse button, and the display will re-draw according to the new scale.

Zooming Out on the Display

To zoom out and see the entire SimGel, click on the display using the right mouse button.

Formatting the Display

To change the display format, click on the **Format** button. This activates the **Format** screen. Here you will find controls for changing the style and color of the electropherograms, the color gradient for SimGel and the option to normalize the plots.

To see both electropherograms on the same scale choose normalize from the **Format** screen.

To save your customized settings as the default settings, push the "Save as Default" button.

Printing the SimGel Report

To print the SimGel display, push the button on the main SimGel screen labeled **Print**. This will open a standard Print dialog box. Different printers and print settings can be defined for printing the SimGel report.

Glossary

BioSize Standard Definition Files

BioSize standard definition files are those files that contain migration time and molecular weight information for a protein standard. They are created when a standard definition is saved and have the DOS extension STD.

Bracket Button

The bracket button is used to average the migration time from two different standard runs.

Internal Reference

An internal reference is a known component added to the standard and sample solutions. Its migration time is used to calculate relative migration times. In some cases an internal reference can increase the accuracy of molecular weight determination.

LSQ

Least squares calculation.

Migration Time

Migration time is the time required for a component to travel through the capillary to the detector window.

Mode

There are two modes of operating this software. BioSize mode and SimGel mode. BioSize is used to estimate molecular weights, and SimGel is used to simulate gels.

Out-of-Range Points

Points that are outside the molecular weight range defined by the protein standard.

Protein Standard

A series of proteins whose molecular weights are known.

Reference Peak

The reference peak is the peak corresponding to the internal reference. It is used to calculate relative migration of sample proteins. When used in an analysis, all migration times are related to the migration time of the internal reference.

Reference Window

The reference window determines the time range the software will search for a reference peak when it analyzes data.

Standard Run

A standard run is one based on the analysis of standard proteins.

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